

Measuring Planck's Constant using Photoelectric Effect

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1. RESULTS AND DISCUSSION

The experiment result shows that the stop voltage is depend on the wavelength (or frequency) of the light but is independent on the intensity of the light, which is matches the Quantum Model. The data consists of 5 light colors and 2 orders of the spectrum, to vary the wavelength and the intensity of the light respectively.

The h measured in this experiment are $(6.35509 \pm 0.00626) \times 10^{-34} J \cdot s$ and $(6.84747 \pm 0.00626) \times 10^{-34} J \cdot s$.

The W_0 are $1.38645J$ and $1.63812J$. The uncertainty of W_0 is $4.2168 \times 10^{-22} J$.

1.1. Measurement and Data

Light	Wavelength	Frequency	V_s
1^{st} order	nm	Hz	V
UV	365.5	8.208×10^{14}	1.837 ± 0.001
Vio	404.7	7.413×10^{14}	1.556 ± 0.001
Blue	435.8	6.884×10^{14}	1.422 ± 0.001
Green	546.1	5.493×10^{14}	0.760 ± 0.001
Yellow	578.0	5.190×10^{14}	0.675 ± 0.001
2^{nd} order	Wavelength	Frequency	V_s
UV	365.5	8.208×10^{14}	1.802 ± 0.001
Vio	404.7	7.413×10^{14}	1.566 ± 0.001
Blue	435.8	6.884×10^{14}	1.403 ± 0.001
Green	546.1	5.493×10^{14}	0.746 ± 0.001
Yellow	578.0	5.190×10^{14}	0.506 ± 0.001

NOTE—There is no uncertainty for Δt given in the ninth trial of this experiment, since two measurements were giving the same value.

As Figure 1 and Figure 2 show, the regression line slopes are nearly consistent($1^{st} = 3.97193 \times 10^{-15}$, $2^{nd} = 4.27967 \times 10^{-15}$) even though the order of the spectrum is adjusted from 1 to 2. Figure 1 shows the 1^{st} spectrume, which has a higher intensity, but the stop voltage is consistent with the 2^{nd} (Figure2) spectrum, which has a lower intensity.

1.2. The Theory and Experiment Result

Frequency vs.Stop Voltage

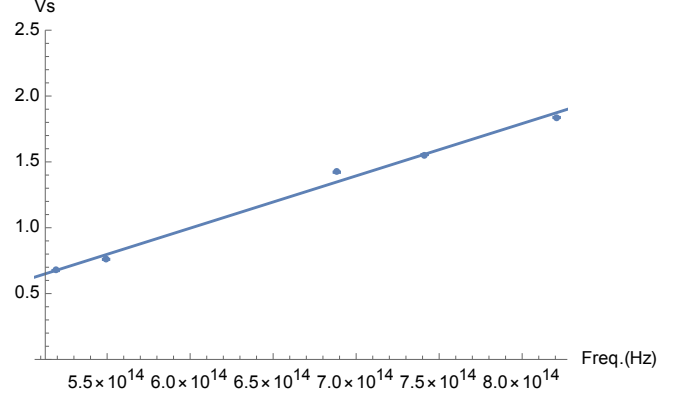


Figure 1. Result of least square fit of the first order spectrum data. The best-fit slope is 3.97193×10^{-15} with uncertainty 3.91291×10^{-18} . The best-fit intercept is -1.38645 with uncertainty 0.0026355 . There are error bars along y-axis, but they are very small and not clearly reflected in this plot.

Frequency vs.Stop Voltage

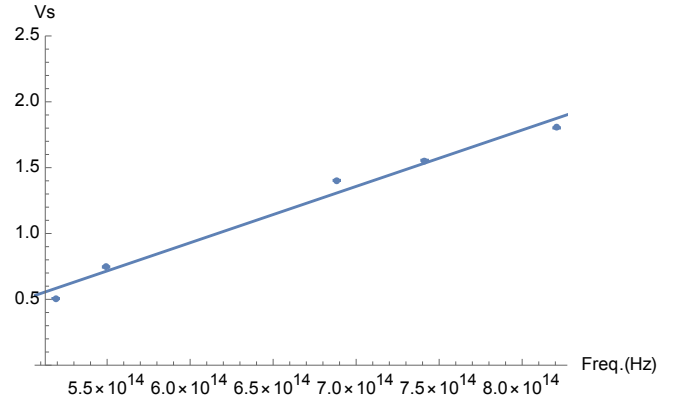


Figure 2. Result of least square fit of the second order spectrum data. The best-fit slope is 4.27967×10^{-15} with uncertainty 3.91291×10^{-18} . The best-fit intercept is -1.63812 with uncertainty 0.0026355 . There are error bars along y-axis, but they are very small and not clearly reflected in this plot.

The experimental result indicates that the intensity in the different order spectrum does not affect the stop

voltage of the corresponding wavelength or frequency. Theoretically, the stop voltage (KE_{max}) should not change either because, according to the quantum model, h is a constant, and therefore h/e is also a constant which is the slope of the regression line. In the experimental measurement model, the slope of the line is h/e , which is a constant. The two coincide: the energy of light is independent of the light intensity and depends on light frequency.

$$E = hf = eV + W_0$$

$$V_s = hf/e - W_0/e$$

1.3. Uncertainty Propagation and Analysis

$$\sigma_f^2 = \sigma_x^2 \left(\frac{\partial f}{\partial x} \right)^2$$

$$m = \frac{1}{e}h$$

$$\sigma_m = \sigma_h \frac{\partial h}{\partial m} = \sigma_h \left(\frac{1}{e}h \right)' = \sigma_h \frac{1}{e}$$

$$\sigma_h = \sigma_m \times e$$

The uncertainty of the slope $\sigma_m = 3.91291 \times 10^{-18}$ is given by the least square fit. Then, $\sigma_h = 0.00626$ is obtained by $\sigma_h = \sigma_m \times e = (3.91291 \times 10^{-18}) \times (1.602 \times 10^{-19})$

Therefore, the complete uncertainty expression of h are $(6.35509 \pm 0.00626) \times 10^{-34} J \cdot s$ for Figure 1 and $(6.84747 \pm 0.00626) \times 10^{-34} J \cdot s$ for Figure 2.

The uncertainty may come from the detector's sensitivity, the cable connection between the multi-meter and the apparatus's output, the power battery's status, the clearance of the color stripes. According to the model and experiment design, the mainly uncertainty comes from the random error when measuring the frequency of the light (dependent variable x). The random error arises at not effectively separating the different light stripes, resulting in the detector collecting different light frequencies. For example, when measuring violet light, likely, the adjacent UV light that needs to be blocked is also collected into the meter.

1.4. Limitation

Experiment data only consists of two orders of spectrum. Higher order spectrum is awaiting for further measurement.